STAT 385 Homework 3 Fall 2026

Due: September 18, 2026

Instructions & Setup:

- The required datasets for this assignment are available in the `ISLR2` R package. To load the environment and access the datasets, run the following setup code in R:

```
install.packages("ISLR2")  # Run once if not already installed
library("ISLR2")
# Datasets used in this homework: Carseats, Boston
```

Question 1: Multiple Linear Regression on `Carseats` [Total: 37 pts]

This problem uses the `Carseats` dataset from the `ISLR2` package. Your objective is to predict child car seat sales (`Sales`) using demographics and qualitative indicators.

(a) **[5 pts]** Fit a multiple linear regression model predicting `Sales` using the following three predictors: `Population`, `Income`, and `US`. Display the model summary.

(b) **[4 pts]** Provide a clear, detailed interpretation of each coefficient in the fitted model. *(Hint: Pay close attention to qualitative/categorical variables like `US` and state what the baseline reference level is.)*

(c) **[6 pts]** Write out the complete fitted regression equation using mathematical notation. Explicitly define any dummy/indicator variables created for qualitative factors.

(d) **[3 pts]** Based on the regression summary, for which predictor variables can you reject the null hypothesis $H_0 : \beta_j = 0$? Use a significance level of $\alpha = 0.05$.

(e) **[5 pts]** Based on your findings in part (d), fit a smaller, reduced multiple regression model using **only** the predictors that showed statistically significant evidence of an association with `Sales`.

(f) **[8 pts]** Compare the model fit between the full model in (a) and the reduced model in (e). Discuss relevant metrics such as R^2, Adjusted R^2, Residual Standard Error (RSE), and the overall F-statistic. Which model is preferred and why?

(g) **[6 pts]** Using your reduced model from part (e), calculate and report 95% confidence intervals for all regression coefficient(s).

Question 2: Simulation Study & Simple Linear Regression [Total: 47 pts]

In this exercise, you will create a synthetic dataset to explore the empirical properties of simple linear regression, sample variability, and the effect of random error noise.

Important Note on Reproducibility: Set the random seed to `2025` immediately before generating your data in part (a) by executing:

```
set.seed(2025)
```

(a) **[2 pts]** Using the `rnorm()` function, generate a vector x containing $N = 150$ observations drawn from a standard normal distribution $\mathcal{N}(0, 1)$. This vector represents feature X.

(b) **[2 pts]** Using `rnorm()`, generate a vector ϵ (epsilon) containing 150 error terms drawn from a normal distribution $\mathcal{N}(0, 0.36)$ with mean $\mu = 0$ and variance $\sigma^2 = 0.36$ (i.e., standard deviation $\sigma = 0.6$).

(c) **[3 pts]** Using x and ϵ, generate the response vector y according to the true data-generating linear model:

$$Y = 5 + 0.8X + \epsilon \tag{3.39}$$

What is the length of the resulting vector y? State the exact values of the true population intercept (β_0) and slope (β_1).

(d) **[4 pts]** Construct a scatterplot displaying the relationship between x and y. Describe any patterns, direction, or spread you observe in the data.

(e) **[6 pts]** Fit a least-squares simple linear regression model to predict y using x. Report the estimated parameters $\hat{\beta}_0$ and $\hat{\beta}_1$. How do these estimated values compare to the true population parameters β_0 and β_1?

(f) **[6 pts]** Display the scatterplot from part (d) again. Overlay two lines:

i. The estimated least-squares regression line (e.g., using `abline()`).

ii. The true population regression line ($Y = 5 + 0.8X$) using a distinct color and line style.

Include a clear legend via `legend()` distinguishing the two lines.

(g) **[4 pts]** Fit a polynomial regression model that predicts y using both x and x^2. Is there statistical evidence that the quadratic term x^2 improves the model fit? Explain your reasoning using the p-value.

(h) **[10 pts] Noise Reduction Analysis:** Repeat parts (a) through (f) after modifying the data-generating process to reduce the noise level in the data (keep the linear model coefficients $Y = 5 + 0.8X$ unchanged). *To do this, decrease the variance of the normal distribution used to generate ϵ in part (b) (e.g., set $\sigma^2 = 0.04$).* Describe how reducing the noise impacts the scatterplot, parameter estimates, and overall fit.

(i) **[10 pts]** Calculate and report the 95% confidence intervals for β_0 and β_1 for both the original dataset (part e) and the lower-noise dataset (part h). Compare the widths of these confidence intervals and explain the theoretical reason for any differences.

Question 3: Predicting Crime Rates in the `Boston` Dataset [Total: 39 pts]

This problem explores the `Boston` dataset from the `ISLR2` package. Your goal is to predict the per capita crime rate (`crim`, the response variable Y) using various urban and demographic predictors.

(a) **[16 pts]** For each of the following predictors separately—`nox` (nitrogen oxides concentration), `tax` (property tax rate), `dis` (weighted distances to employment centers), and `ptratio` (pupil-teacher ratio):

- Fit a simple linear regression model to predict `crim`.
- Summarize your findings: in which models is there a statistically significant relationship with the response?
- Produce scatterplots with fitted regression lines to visually support your conclusions.

(b) **[5 pts]** Fit a multiple linear regression model to predict `crim` using **all** available predictors in the dataset. Summarize the results. For which predictors can we reject the null hypothesis $H_0 : \beta_j = 0$ at $\alpha = 0.05$?

(c) **[6 pts]** Compare your findings from the simple linear regression models in part (a) to the multiple linear regression results in part (b):

- Do any coefficients change significantly in magnitude, sign, or statistical significance?
- Explain why simple and multiple regression results can differ (e.g., collinearity or confounding variables).

(d) **[12 pts]** Test for non-linear relationships between the predictors and the response. For each predictor X, fit a cubic polynomial model of the form:

$$Y = \beta_0 + \beta_1 X + \beta_2 X^2 + \beta_3 X^3 + \epsilon$$

Is there evidence of non-linear association for any of the predictors? Justify your conclusions using hypotheses tests (p-values) for the non-linear terms (X^2 and X^3).